

APP DEV 1 PROJECT REPORT

Placement Portal Application

1. Student Details

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About Me:

I am a B.Sc. (Hons) Electronics student from Delhi University with a strong interest in web development, AI, robotics, and full-stack application design. I enjoy building practical systems that solve real-world workflow problems, such as student placement management and recruitment automation.

2. Project Details

Project Title: Placement Portal Application

Problem Statement:

To design and develop a web-based placement portal that enables Admin, Company, and Student users to manage the complete campus recruitment workflow, including company approvals, placement drive creation, student applications, status tracking, and placement history.

Approach:

The project was built using Flask with a modular structure based on role-specific dashboards for Admin, Company, and Student. SQLite was used as the database with SQLAlchemy ORM for programmatic schema creation. Jinja2 templates and Bootstrap were used for a responsive front-end, while Flask-Login handled authentication and role-based access control.

3. AI / LLM Declaration

I used ChatGPT (GPT-5) to assist in project planning, Flask route structuring, debugging, GitHub workflow, and documentation formatting.

The extent of AI/LLM usage is approximately 15–20%, limited to architecture guidance, debugging support.

All final implementation, testing, integration, debugging on the local machine, GitHub milestone commits, and deployment workflow were completed manually by me.

4. Technologies and Frameworks Used

Technology	Purpose
Flask	Core backend framework
Flask-Login	Authentication and session management
SQLAlchemy	ORM for SQLite database
SQLite	Local database storage
Jinja2	Dynamic HTML template rendering
HTML/CSS	Front-end structure and styling
Bootstrap 5	Responsive UI design
Werkzeug Security	Password hashing
Git + GitHub	Version control and milestone tracking

5. Database Schema / ER Design

Tables Used:

1. User: id, name, email, password, role, approved, active, blacklisted, phone, degree, cgpa, resume
2. CompanyProfile: id, user_id, company_name, hr_contact, website, approval_status
3. Drive: id, company_id, title, description, eligibility, deadline, status
4. Application: id, student_id, drive_id, application_date, status

Relationships:

User (Company) → One-to-Many → Drive

User (Student) → One-to-Many → Application

Drive → One-to-Many → Application

6. Architecture and Features

Architecture:

app.py → routes and controller logic

models.py → database models

config.py → configuration

templates/ → Jinja2 pages

static/ → CSS files

placement.db → SQLite database

Implemented Features:

- Predefined Admin superuser
- Company registration with admin approval
- Student self-registration and profile update
- Role-based dashboards
- Placement drive workflow
- Resume upload
- Duplicate application prevention
- Status tracking and history
- Blacklisting and deactivation

7. Video Presentation

Drive Link:

<https://drive.google.com/file/d/1jS666WPx9NSTVxbK7Qe7DQVgo8EbsLJ9/view?usp=sharing>